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Abundances and distribution of minerals and elements in high-alumina coal fly ash from the Jungar Power Plant, Inner Mongolia, China

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ABSTRACT

The fly ash from the Jungar Power Plant, Inner Mongolia, China, is unique because it is highly enriched in alumina (Al_2O_3>50%). The fly ash mainly consists of amorphous glass and mullite and trace amounts of corundum, quartz, char, calcite, K-feldspar, clay minerals, and Fe-bearing minerals. The mullite content in fly ash is as high as 37.4% because of high boehmite and kaolinite contents in feed coal. Corundum is a characteristic mineral formed during the combustion of boehmite-rich coal.

Samples from the economizer were sieved into six size fractions (<120, 120–160, 160–300, 300–360, 360–500, and >500 mesh) and separated into magnetic, mullite + corundum + quartz (MCQ) and glass phases for mineralogical and chemical analysis. The corundum content increases but amorphous glass decreases with decreasing particle size. Fractions of small particle sizes are relatively high in mullite, probably because mullite was formed from fine clay mineral particles under high-temperature combustion condition. Similarly, fine corundum crystals formed in the boiler from boehmite in feed coal. The magnetic phase consists of hematite, magnetite, magnesioferrite, and $MgFeAlO_4$ crystals. The MCQ phase is composed of 89% mullite, 6.1% corundum, 4.5% quartz, and 0.5% K-feldspar.

Overall, the fly ash from the power plant is significantly enriched in Al_2O_3 with an average of 51.9%, but poor in SiO_2, Fe_2O_3, CaO, MgO, Na_2O, P_2O_5, and As. Arsenic, TiO_2, Th, Al_2O_3, Bi, La, Ga, Ni, and V are high in mullite, and the magnetic matter is enriched in Fe_2O_3, CaO, MnO, TiO_2, Cs, Co, As, Cd, Ba, Ni, Sb, MgO, Zn, and V. The remaining elements are high in the glass fraction. The concentration of K_2O, Na_2O, P_2O_5, Nb, Cr, Ta, U, W, Rb, and Ni do not clearly vary with particle size, while SiO_2 and Hg decrease and the remaining elements clearly increase with decreasing particle size.

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1. Introduction

According to the report released by Coal Industrial Society of China in January 2009, China produced a total of about 2.716 billion tons of coal in 2008, 7.65% higher than last year. Chinese power plants consumed 1.09 billion tons of coal and generated 81.8% of the total electricity in 2005 (Zhou, 2006). The demand for coal as a major energy resource in China is projected to increase in the near future. Meanwhile, 293 million tons of coal combustion residues were generated by Chinese power plants in 2005 (Zhou, 2006) and the amount of that will increase every year.

Environmental problems can be caused by coal mining, transportation, storage and utilization, among which coal combustion is most significant (Meij and te Winkel, 2009). Toxic elements released during coal combustion have influenced normal growth of animals and vegetation, and human health in some countries and areas (Finkelman et al., 2002; Dai et al., 2005). On the other hand, certain elements are concentrated in fly ash that can be potentially of industrial value (Seredin, 1996; Seredin and Finkelman, 2008). Vassilev and Vassileva (2007) indicated that the abundance and modes of element occurrence play a leading role in specification and utilization of fly ash. Trace-element partitioning in pulverized-coal boilers is primarily dependent on the feed coal, boiler types, and pollution control system operating conditions (Guedes et al., 2008; Mardon et al., 2008; Huggins and Goodarzi, 2009).

Minerals are major carriers of elements in coal combustion residues. Minerals in coal combustion residues can be primary, secondary and tertiary in origin (Vassilev and Vassileva, 1996). Research on mineral behaviors during coal combustion can provide useful information for boilers to run safely and economically.

The purpose of this paper is to determine the mineralogical and chemical compositions of high-alumina fly ash generated at the Jungar Power Plant, Inner Mongolia, China. Both the coals from the Jungar

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coalfield and the fly ash from the Jungar Power Plant are unique because the raw coals are highly enriched in boehmite (Dai et al., 2006a, 2008) and the fly ash is high in alumina (Al_2O_3>50%). Because of the discovery of high Al and Ga contents in both coal and fly ash, the management of the Heidaigou Surface Mine, which is owned by Shenhua Group Zhungeer Energy Corporation Limited, has financially supported us to conduct a research project to investigate the modes of occurrence of Ga and Al in coals and to develop technology to extract the metals from associated combustion residues.

2. Samples and methods

The location of the Junger Coalfield was described by Dai et al. (2006a). Feed coals, fly and bottom ashes of the Jungar Power Plant were sampled in a period of 5 days (from June 20 to 24, 2006). Samples of pristine fly ash were collected from the economizer of the power plant with a special sampling device because the flue gas along with fly ash was disposed through wet ash concentration units at this power plant. Samples of wet fly ash (WA) were also collected at the ash concentration units and bottom ash was collected at the bottom hoppers.

Fly ashes from economizer were screened using sieves of 120 (120 μm), 160 (96 μm), 300 (48 μm), 360 (41 μm), and 500 mesh (25 μm). Wet fly ash was not used because of its pozzolanic nature, and even after being dried, particles could be conglutinated with each other. In addition, since it had reacted with water in the ash concentration units, some new minerals may be formed in wet fly ash, such as calcite formed from lime and gypsum formed from anhydrite.

All the feed coals were analyzed for moisture, ash yield, and sulfur content. Loss on ignition of fly ash was determined by heating the sample at 1050 °C for at least 1 h. Epoxy-bound pellets were made from fly ash samples and prepared to a polish with 0.05-μm alumina.

The minerals in fly ash samples were quantified by powder X-ray diffraction method, based on the method described in Petroleum and Gas Standard of China (SY/T 6210-1996: Quantification methods of total clay and other minerals in sedimentary rocks). The XRD pattern was recorded over a 2θ interval of 3°–70°, with a step size of 0.02°. A scanning electron microscope equipped with an energy-dispersive X-ray spectrometer was used to study the surface characteristics and determine the distribution of elements in fly ash. The accelerating voltage was 20 KV and the beam current was 10^{-10} A. X-ray fluorescence spectrometry (XRF) was used to determine the oxides of major elements, including SiO_2, Al_2O_3, CaO, K_2O, Na_2O, Fe_2O_3, MnO, MgO, TiO_2, and P_2O_5. Inductively coupled plasma mass spectrometry (ICP-MS) was used to determine trace elements in coal and ash samples that were first subjected to microwave digestion in ultrapure concentrated HNO_3 using high pressure quartz vessels. However, Hg, Se and As were determined by atomic fluorescence spectrometry (AFS) after dissolution by $HNO_3 + HClO_4 + HCl$. Fluorine was determined by pyrohydrolysis in conjunction with a fluoride ion-selective electrode, following the method described in Chinese National Standard GB/T 4633-1997.

In order to study chemical and mineral compositions in different phases of fly ash, the fly ash was separated into magnetic, mullite + corundum + quartz (MCQ), and glass phases. The magnetic materials were first separated by a hand magnet. The nonmagnetic fraction was then treated with 4% HF to dissolve the glass and the solid residues were mainly mullite, quartz, and corundum.

3. Results and discussion

3.1. Properties of feed coal at the power plant

The No. 6 coal (the uppermost Taiyuan Formation of Pennsylvanian age) of the Jungar Coalfield, Inner Mongolia, has a thickness between 2.7 and 50 m and an average thickness of 30 m (Dai et al., 2006a, 2008). The No. 6 coal at the Heidaigou Mine, Jungar Coalfield, is divided into 7 benches which are ZG6-1 to ZG6-7. Petrology and geochemistry of the coals have been reported (Dai et al., 2006a). The vitrinite random reflectance of the No. 6 coal is 0.58% (Dai et al., 2006a). Only mined coal from benches ZG6-5 to ZG6-7 is delivered to the power plant after having been washed. Coal from benches ZG6-1 to ZG6-2 is largely discarded because of a high ash yield. Coal mined from benches ZG6-3 to ZG6-4 is sold to other markets because of good qualities. The No. 6 coal is remarkably high in boehmite (33.7% of the total minerals). The origin of boehmite in coal is related to transportation of aluminum compounds from weathered and oxidized bauxite crust into the basin during peat deposition (Dai et al., 2006a).

The Jungar Power Plant generates about 200 MW of power and 380 Kt of fly and bottom ashes every year. The fly ash collected by ash concentration units accounts for 90% of total combustion residues which are piped into a landfill 3 km away to the northeast of the power plant.

Table 1 is a list of moisture, ash yield, and sulfur content in samples (C20–C25) of feed coal collected daily at the power plant in a period of 5 days and the average values of coal from the Heidaigou Mine. The feed coal of the power plant is low-sulfur and high volatile bituminous in rank (the average volatile matter is 33.5% on a dry and ash-free basis; Dai et al., 2006a) from the Heidaigou Surface Mine in the Jungar Coalfield. No significant variation of these characteristics is noticed between individual feed coals. The average ash yield of feed coal at the power plant is 33.0%, which is much higher than the average ash yield of coals from the Heidaigou Mine (17.2%) because the power plant uses the middling from the preparation plant from the lower benches (ZG6-5 to ZG6-7) of coals only from the mine.

Although the most boehmite-rich coals (ZG6-3 to ZG6-4) were excluded from the preparation plant, boehmite content in feed coal is still as high as 21.1% (mineral matter basis). Other minerals in feed coal are mainly 71.1% kaolinite, 3% gypsum, 2.5% calcite, and 1.9% quartz (mineral matter basis; Chen, 2005). The raw coal is enriched in aluminum and gallium (Dai et al., 2006a,b, 2008). Consequently, Al and Ga are highly concentrated in the fly ash (Dai et al., 2006b; Zhang et al., 2006).

3.2. Minerals and amorphous glass in fly and bottom ashes

3.2.1. Minerals in pristine bottom and fly ashes

As in most cases, both fly ash and bottom ash contain an important proportion of amorphous aluminosilicate glass, minerals, and char (Table 2). Fly ashes from the economizer and wet ash concentration units have much lower loss on ignition than the bottom ash. Minerals in bottom and fly ashes identified by XRD include mullite, corundum, quartz, and minor amounts of calcite and K-feldspar. The mullite and corundum contents in bottom ashes are much lower than those in fly ash. The dry fly ash from the economizer has a lower mullite and a higher glass content than the wet ashes from concentration units (Table 2). Fly ashes are composed of spherical, semispherical, and

Table 1
Moisture, ash yield, and sulfur content of feed coal from the Jungar Power Plant (wt.%).

Samples	M_{ad}	A_d	$S_{t,d}$
C20	3.62	36.9	0.33
C21	3.51	34.1	0.41
C22	3.35	33.2	0.42
C23	3.13	30.5	0.43
C24	2.91	30.2	0.39
Average feed coal	3.3	33.0	0.4
Heidaigou Mine[a]	5.19	17.7	0.73

M, moisture; ad, as-determined; A, ash; d, on a dry basis; S_t, total sulfur. C20–C24, feed coal collected daily for a period of 5 days.
[a] From Dai et al. (2006a).

Table 2
Mineral and glass abundances and loss on ignition (LOI) of coal combustion residues from the Jungar Power Plant (in wt.%; samples collected on June 21, 2006).

Samples	Minerals					Glass	LOI
	Mullite	Corundum	Quartz	Calcite	K-feldspar		
Fly ash (w)	37.4	3.9	2	0.2	0.2	52.6	2.86
Fly ash (e)	34.9	4	1.6	0.3	0.2	54.8	3.15
Bottom ash	27.2	3.2	2	0.5	0.3	54.4	12.0

Fly ash (w), fly ash collected from the wet ash concentration unit; Fly ash (e), fly ash from the economizer.

irregular grains of different sizes as observed under electron scanning microscope (Fig. 1).

3.2.1.1. Mullite. Mullite is a typical mineral in most fly ashes formed during coal combustion in power plant. The source of mullite in fly ash is mainly kaolinite in raw coal (Raask, 1985; Spears, 2000). The high mullite content in the fly ash (34.9–37.4%) is related to a high kaolinite content in the feed coal (71.1% of the mineral matter is kaolinite).

The lower mullite content in bottom ash than in both dry and wet fly ashes may indicate that bottom ash experienced a relatively shorter time at high temperature than fly ashes. Needle mullite crystallites on glass sphere (Fig. 2) indicate that mullite crystallizes from amorphous aluminosilicate melt during combustion but not having transformed directly from kaolinite. The various spherical shapes of many fly ash particles suggest that they solidified from a viscous fluid state (Henry et al., 2004). Needle mullite crystallites of varied orientation (Fig. 2) are capable of raising the strength of cenosphere shells (Sokol et al., 2000).

3.2.1.2. Corundum. Corundum is generally rare in both raw coal and combustion residues, although it can be found in some coals as a accessory minor mineral (Vassilev et al., 1994), high-temperature ashes (Ward, 2002), and coal combustion residues (Vassilev et al., 2003, 2005). Corundum does not occur in feed coals of the Jungar Power Plant. Corundum in fly ash may have crystallized from boehmite in feed coal during coal combustion and thus is a characteristic mineral formed during the combustion.

3.2.1.3. Other minerals. Quartz is a common mineral in ordinary fly ash; however, owing to the low quartz content in feed coal, fly ash from the Jungar Power Plant contains a relatively low content of quartz, less than 2%. Quartz may be primary in origin because of its high melting point (1713 °C). It could also be secondary in origin,

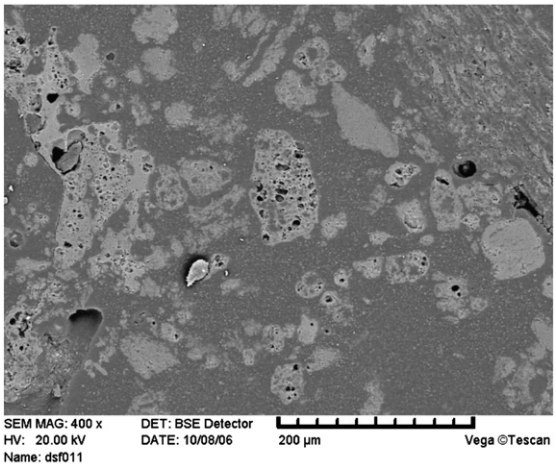


Fig. 1. A broad view of fly ash from Jungar Power Plant (SEM, back-scattered electron image on a polished section).

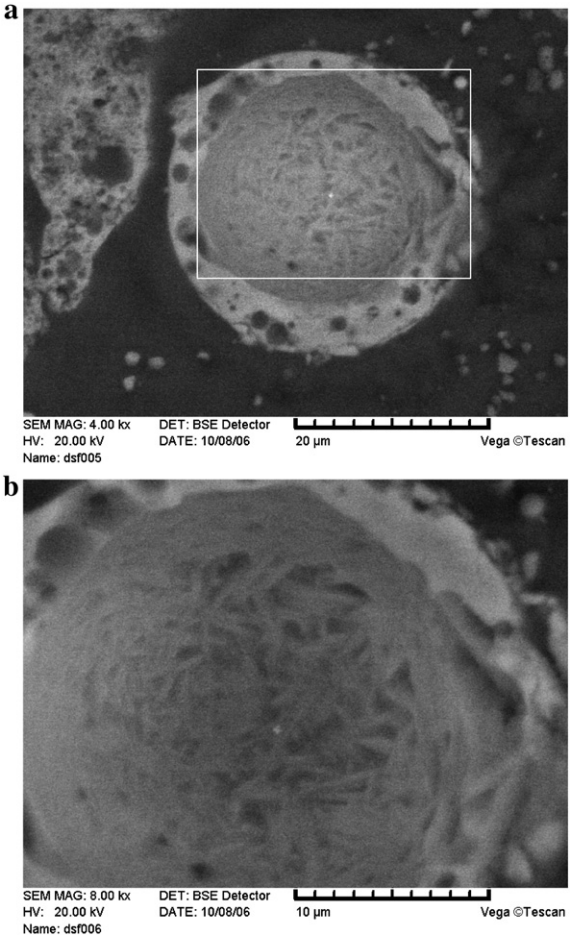


Fig. 2. Needle mullite crystallites (SEM, back-scattered electron image on polished sections). (a) A solid aluminosilicate sphere. (b) Needle mullite crystals on the surface of the sphere, marked area in A is enlarged.

formed from the SiO_2–Al_2O_3 system (Vassilev and Vassileva, 1996). Quartz in the dry fly ash from the economizer is slightly lower than that in the wet ash from the ash concentration units (Table 2), which might be related to continued quartz crystallization from the melt.

Minor amounts of calcite and K-feldspar are found in Jungar fly ash. There is a small amount of calcite in raw coal (0 to 1.1%) (Dai et al., 2006a), but it should be decomposed during combustion. Calcite in fly ash may be a secondary mineral formed from CaO at a low temperature.

3.2.1.4. Amorphous glass phase. Glass is the most abundant constituent in fly and bottom ashes. It is 54.8, 54.4, and 52.6% in fly ash from the economizer, bottom ash, and wet ash, respectively (Table 2). Unlike the report of Mardon and Hower (2004) indicating that the percentage of glass in fly ash increases from the economizer to the electrostatic precipitator, the percentage of glass in fly ash of the Jungar Power Plant decreases slightly from the economizer to the wet ash concentration unit. It may be related to the considerable mullite content in wet fly ash of the Jungar Power Plant as compared with other power plants. Mullite is derived from the glass as discussed above.

3.2.2. Mineralogical composition in size fractions of fly ash

The proportions of six size fractions of the dry fly ash and their mineral compositions are listed in Table 3. A total of 81.2% of the fly ash passed 160 mesh and 40.7% passed 300 mesh sieve (Table 3). The 160–300 mesh size is the major size fraction (40.5%) of the Jungar Power Plant.

Table 3
Glass and mineral contents in six size fractions of dry fly ash from the Jungar Power Plant (in wt.%).

Particle size (mesh)	Proportion	Minerals						Glass
		Mullite	Corundum	Quartz	K-feldspar	Calcite	Clay minerals	
<120	8.7	32.3	3.5	1.4	0.1	bdl	bdl	51.5
120–160	10.1	35.4	4.5	1.1	bdl	bdl	bdl	53.4
160–300	40.5	35.3	4.8	1.5	0.1	0.2	2	49.4
300–360	9.0	37.5	6.4	bdl	0.1	bdl	bdl	49.7
360–500	16.0	35.9	7.9	1.5	0.3	0.3	2.6	45.9
>500	15.7	35.6	10.9	0.9	0.1	bdl	1.3	45

bdl, below detection limit.

Table 4
Inorganic phases in magnetic and non-magnetic fly ash fractions from Jungar Power Plant (wt.%).

Sample	Minerals							Amorphous material
	Mullite	Quartz	Corundum	Feldspar	Calcite	Hematite	Magnetite + magnesioferrite + $MgFeAlO_4$	
Magnetic fly ash	25.1	2.5	2.3	0.8	0.4	13	1.5	54.4
Nonmagnetic fly ash	38.1	2.1	4	bdl	bdl	bdl	bdl	55.8
MCQ phase	89	4.5	6.1	0.5	bdl	bdl	bdl	bdl

Some studies showed that the mineral composition greatly varies in different size fractions and the smaller size fractions usually contain higher content of glass than the larger ones (Matsunaga et al., 2002; Qian, 2002; Mardon and Hower, 2004). A study by Xu (2004) indicated that the content of mullite and quartz decreases but glass increases with decreasing particle size of fly ash. However, the data from this study are different from Qian (2002), Matsunaga et al. (2002), Mardon and Hower (2004), and Xu (2004). The semiquantitative XRD results indicate that the glass content decreases but mullite increases with decreasing particle size of fly ash. It seems that the smaller size of fly ash from the Jungar Power Plant is more favorable for the mullite formation. This is consistent with the report by Goodarzi (2006a,b) that the mullite content of fly ash collected by the baghouse, which has a greater ability to capture fine fly ash particles than ESP, is greater than the ESP ash from the same station, indicating that mullite is concentrated in fine fly ash particles. The fly ash is very low in quartz, K-feldspar, clay minerals, and calcite. Mullite and glass are dominant phases in fly ash; the glass content is low when mullite content is high in fly ash.

The corundum content increases as the fly ash particle size decreases; it is particularly high in the <500 mesh fraction (10.9%). Corundum crystals probably formed in the boiler from boehmite while most of boehmite remains as glass because of short residence time in the boiler.

The quartz content in fly ash of different size fraction is low and ranges between below detection limit to 1.5%, with a weak relationship with size distribution. Clay minerals were determined in the 160–300 mesh and two smallest size fractions. Calcite was determined in two size fractions (160–300 and 360–500 mesh).

3.2.3. Minerals in magnetic and nonmagnetic fractions in fly ash

The proportion of magnetic fly ash is generally low, ranging from 0.5 to 18.1% (Bibby, 1977). The magnetic fly ash accounts for about 1.8% of the total fly ash from the Jungar Power Plant. Table 4 listed the phase compositions of magnetic and nonmagnetic fly ash fractions from Jungar Power Plant. The magnetic fly ash from Jungar is composed mainly of hematite, magnetite, magnesioferrite, and $MgFeAlO_4$ crystals. Due to the low content of magnesioferrite and $MgFeAlO_4$ crystal and their diffraction peaks overlapping with that of magnetite, the content of magnesioferrite and $MgFeAlO_4$ crystal were merged into the magnetite content in calculation.

The feed coals for the Jungar Power Plant contain Fe-bearing minerals such as pyrite and siderite (Dai et al., 2006a). Thus, the Fe-bearing minerals in the Jungar fly ash were probably derived from pyrite and siderite in raw coal.

The combustion temperature is an important factor that influences the types and contents of Fe-containing minerals in the fly ash. With increasing temperature, Fe_3O_4 is first oxidized to the solid solution of $\gamma 2Fe_2O_3$ or a mixture of $\gamma 2Fe_2O_3$ and Fe_3O_4, and then transforms into $\alpha 2Fe_2O_3$. When the temperature reaches 1400 °C, Fe_2O_3 transforms into Fe_3O_4 (Sun et al., 2005). The major magnetic mineral in the Jungar Powder Plant is hematite, accounting for 13%, much higher than that of magnetite (1.5%), indicating that the combustion temperature is less than 1400 °C, which is consistent with the finding of Kukier et al. (2003) that predominance of magnetite over hematite indicates that combustion temperature exceeded 1388 °C. The nonmagnetic hematite was classified into the magnetic fraction not only because of its high Fe content, but also due to the fact that nonmagnetic hematite may bear magnetism after being heated at a high temperature.

The Jungar magnetic fly ash contains a 54.4% noncrystalline matter (Table 4). Zhao et al. (2006) separated the magnetic spheres into four types: Fe-oxide phase (Fe≥75%), aluminosilicate-containing Fe-oxide (75%>Fe≥50%), Fe-rich aluminosilicate-oxide (50%>Fe≥25%), and Fe-containing aluminosilicate-oxide (Fe<25%). The semiquantitative SEM-EDX results showed that most of the Jungar magnetic glass in the fly ash is Fe-containing aluminosilicate-oxides. The Jungar magnetic fly ash also contains 25.1% of mullite and trace amounts of corundum, quartz, plagioclase and calcite. This is attributed to the conglutination of minerals in the boiler during combustion.

The occurrence modes of ferrospheres include plerosphere (Fig. 3a–c), ferri cenosphere (Fig. 3d), porous microsphere (Fig. 3e), and dense ferrosphere (Fig. 3f). Minerals on the surface of the ferrospheres are generally hematite and magnetite. However, the degree of crystallization of the minerals on the surface is different. The minerals with poor crystal shapes indicate rapid quenching, and the minerals with relatively well-developed crystal shapes generally occur in dendritic (Fig. 3b), skeletal (Fig. 3c), and grain (Fig. 3g) forms.

The nonmagnetic fraction of fly ash comprises 38.1% mullite, 55.8% amorphous glass and trace amounts of quartz and corundum. However, the amorphous glass and crystals in the nonmagnetic fly ash can not be separated from each other using a hand magnet because the crystals are distributed in the glass in the nanometer scale under transmission electron microscope although there is a clear interface between crystals and glasses under transmission electron microscope (Qian et al., 1988). The better way to separate the glass from non-magnetic crystals of nonmagnetic fly ash is by chemical method.

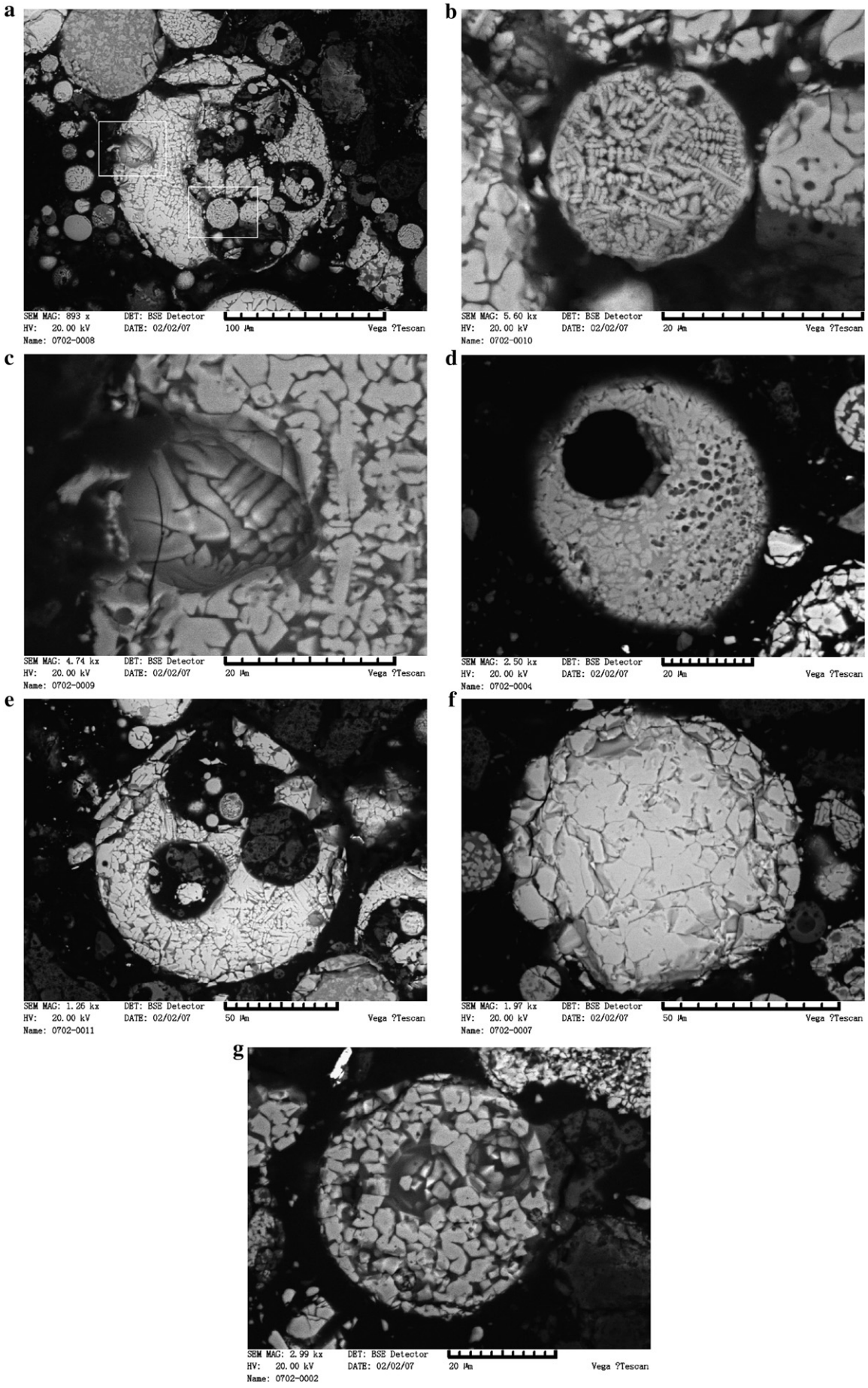

a
SEM MAG: 893 x DET: BSE Detector
HV: 20.00 kV DATE: 02/02/07 100 μm Vega ?Tescan
Name: 0702-0008
b
SEM MAG: 5.60 kx DET: BSE Detector
HV: 20.00 kV DATE: 02/02/07 20 μm Vega ?Tescan
Name: 0702-0010
c
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d
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g
SEM MAG: 2.99 kx DET: BSE Detector
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Name: 0702-0002

Table 5
Inorganic phases in fly ash from Jungar Power Plant (wt.%).

Phases	Proportions
Magnetic phase	1.8
MCQ phase	43.4
Glass phase	54.8

Acids can be used to dissolve the amorphous glasses (Henry et al., 2004) and solid mineral phases composed of mullite, corundum, and quartz will be obtained. The acids can be hydrofluoric acid, hydrochloric acid, acetic acid, and oxalic acid (Qian, 2002). For example, Henry et al. (2004) used the mixture of 15% HF and 25% HNO_3 to separate the glass and crystals in the nonmagnetic fraction. Hulett et al. (1980) dissolved the nonmagnetic fly ash and found that the residues were dominated by mullite occurring as frameworks under scanning electron microscope; and it was deduced that the glasses occurred as fillings within the mullite frameworks before dissolution.

Owing to the low contents of quartz (2.1%) and corundum (2.1%) in the nonmagnetic fly ash and the very low dissolution speed of the two minerals in diluted HF (Hulett et al., 1980), 4% HF was used to dissolve amorphous glasses in the nonmagnetic fly ash in this study. Thus, three phases including magnetic, mullite + corundum + quartz (MCQ), and glass phases can be separated using methods of magnetic separation and chemical dissolution. The proportions of magnetic, MCQ, and glass fractions were 1.8, 43.4, and 54.8%, respectively (Table 5).

The XRD results of the residues after HF dissolution showed that the MCQ phase consists of 89% mullite, 6.1% corundum, 4.5% quartz, and 0.5% K-feldspar.

3.3. Abundance and distribution of elements in fly ash

3.3.1. Oxides of major elements and trace elements in combustion residues

Table 6 listed the average content of elements in feed coal, dry fly ash, wet fly ash, and bottom ash collected from Jungar Power Plant during June 20 to 24, 2006.

As compared with some reported data (Roy and Griffin, 1982; Moreno et al., 2005; Goodarzi, 2006a,b; Depoi et al., 2008; Mardon et al., 2008; Levandowski and Kalkreuth, 2009), the Jungar combustion residues are significantly enriched in Al_2O_3 but poor in CaO, SiO_2, Fe_2O_3, MgO, As, Cr, Cu, Co, Zn, Ni, V. The average Al_2O_3 content in the wet fly ash from the concentration units and dry fly ash from the economizer is 50.84% and 51.87%, respectively. Two samples of combustion residues from the landfill 3 km away to the northeast of the power plant also have high Al_2O_3 content (51.2% and 50.1%). The Al_2O_3 content is about twice higher than the previous reported data (Roy and Griffin, 1982; Moreno et al., 2005; Goodarzi, 2006a,b; Mardon et al., 2008; Depoi et al., 2008; Levandowski and Kalkreuth, 2009). The ratio of Al_2O_3 to SiO_2 is as high as around 1.3, but those of the reported data (Roy and Griffin, 1982; Moreno et al., 2005; Goodarzi, 2006a,b; Depoi et al., 2008; Mardon et al., 2008; Levandowski and Kalkreuth, 2009) are rather low, around 0.5. The high Al_2O_3 and low SiO_2 in the combustion residues of the Jungar Power Plant are the remarkable characteristics as compared with the ordinary coal combustion residues. The high Al_2O_3 content in fly ash from the Jungar Power Plant is controlled by feed coal in which the mineral matter is mainly composed of kaolinite and boehmite (Dai et al., 2006a).

Table 6
Chemical analysis of elements in fly and bottom ashes from Jungar Power Plant.

Element	Feed coal (μg/g)	Dry fly ash (μg/g)	Wet fly ash (μg/g)	Bottom ash (μg/g)	*f/b*	Re_f	Re_b	Kv (%)
Al_2O_3 (%)	15.64	50.84	51.87	42.52	1.23	1.1	0.9	0
SiO_2 (%)	14.57	39.87	39.92	31.75	1.26	0.9	0.72	7.2
CaO (%)	0.37	1.75	1.33	1.36	0.99	1.23	1.28	0
K_2O (%)	0.29	0.47	0.52	0.37	1.42	0.6	0.43	38.5
TiO_2 (%)	0.83	0.69	0.67	0.77	0.88	0.27	0.31	71.6
Fe_2O_3 (%)	1.19	0.93	0.82	1.22	0.69	0.23	0.34	74.7
MgO (%)	0.05	0.15	0.19	0.2	0.95	1.18	1.28	0
Na_2O (%)	0.04	0.12	0.2	0.22	0.97	1.88	1.98	0
MnO (%)	0.01	0.01	0.01	0.01	1.11	0.53	0.48	45.3
P_2O_5 (%)	0.1	0.09	0.12	0.09	1.33	0.42	0.32	57.4
Li	147	453	413	358	1.15	0.93	0.8	4.2
Be	2.16	5.14	6.1	3.45	1.81	0.94	0.53	5.9
F	493	655	711	273	2.63	0.48	0.19	52.9
Sc	8.34	18.93	23.0	15.0	1.54	0.91	0.59	7.4
V	34.87	68.87	89.5	54.8	1.65	0.85	0.52	14.3
Cr	13.17	39.99	36.9	34.1	1.14	0.94	0.86	2.6
Co	1.39	3.36	5.98	9.7	0.95	1.3	2.28	0
Ni	3.96	11.58	19.0	30.6	0.86	1.57	2.5	0
Cu	17.01	38.86	41.2	27.0	1.54	0.8	0.52	18.6
Zn	58.19	48.79	63.2	36.2	1.79	0.45	0.26	54.4
Ga	24.26	44.37	60.2	43.0	1.42	0.82	0.58	16.5
Ge	1.15	2.94	3.47	2.05	1.71	1	0.59	0
As	0.99	0.98	1.38	0.37	3.72	0.5	0.14	51.1
Se	5.01	2.87	2.52	0.96	3.52	0.16	0.06	83.8
Rb	5.09	11.1	14.4	9.8	1.48	0.96	0.65	2.8
Sr	208.	411	545	313.	1.78	0.87	0.5	12.1
Y	20.49	42.1	54.2	33.8	1.62	0.88	0.55	11.4
Zr	402.	762	1000	599	1.70	0.82	0.49	17.1
Nb	29.1	78.7	89.3	60.6	1.48	1.01	0.69	0
Mo	2.39	3.97	5.02	3.42	1.47	0.69	0.47	29.8
Ag	0.45	0.74	1.03	0.65	1.61	0.77	0.48	22.6
Cd	0.09	0.08	0.2	0.08	3.06	0.75	0.30	26
In	0.09	0.12	0.21	0.07	3.23	0.76	0.24	25.6
Sn	3.28	2.53	8.42	3.45	3.39	0.85	0.35	16.1
Sb	0.28	0.42	0.72	0.25	3.01	0.84	0.30	17.8
Cs	0.34	0.79	0.99	0.71	1.40	1.03	0.74	0
Ba	49.9	124.	152.	125.	1.23	1.00	0.84	0
La	41.2	85.4	104.3	67.3	1.56	0.84	0.54	15.3
Ce	71.77	141	178	117.4	1.52	0.82	0.54	17
Pr	8.1	17.3	21.5	13.9	1.56	0.87	0.56	11.5
Nd	27.6	58.5	72.5	46.8	1.55	0.87	0.56	12.3
Sm	5.24	10.6	13.5	8.66	1.56	0.85	0.55	14.5
Eu	0.92	1.8	2.39	1.47	1.63	0.86	0.53	13.5
Gd	4.67	9.11	11.7	7.52	1.56	0.82	0.53	16.6
Tb	0.7	1.4	1.83	1.13	1.63	0.86	0.53	13.6
Dy	4.22	8.62	10.76	6.94	1.56	0.84	0.54	14.8
Ho	0.83	1.68	2.13	1.34	1.59	0.85	0.54	14.1
Er	2.38	4.92	6.17	3.96	1.57	0.86	0.55	13.3
Tm	0.34	0.7	0.87	0.56	1.55	0.85	0.55	14.1
Yb	2.28	4.77	5.98	3.93	1.53	0.87	0.57	12
Lu	0.32	0.68	0.86	0.55	1.56	0.88	0.57	10.6
Hf	10.7	23.3	27.6	18.9	1.46	0.86	0.59	12.9
Ta	1.87	4.74	5.34	3.83	1.39	0.94	0.68	3.8
W	2.63	15.7	56.7	116	0.77	8.78	18.11	0
Re	0.002	0.004	0.006	0.009	1.16	0.95	2.09	0
Au	0.033	0.06	0.077	0.056	1.367	0.78	0.57	20.8
Hg	0.22	0.07	0.08	0.02	4.617	0.14	0.03	86.1
Tl	0.22	0.45	0.5	0.19	2.68	0.76	0.29	24.8
Pb	35.7	54.1	84.5	22.8	3.71	0.78	0.21	24.2
Bi	0.75	0.86	1.58	0.26	6.05	0.7	0.12	33
Th	24.3	48.1	63.0	40.2	1.57	0.86	0.55	13.5
U	5.38	11.7	14.8	10.3	1.44	0.91	0.63	7.8

f/b = (elemental content in fly ash)/(elemental content in bottom ash).
$Re = ([X]_a/[X]_c) \times (A_{ad}/100)$; $[X]_a$ is the content of elements in fly or bottom ashes; $[X]_c$ is the content of corresponding elements in coal; A_{ad} is the ash yield of the coal; Re_f is Re of fly ash, and Re_b is Re of bottom ash. $Kv = (1 - Re_f \times 0.9/95\% + Re_b \times 0.1) \times 100\%$.

With exceptions of Ca, Fe, Li, Cr, and Se, the contents of most elements in wet fly ash are higher than those in dry fly ash (Table 6).

Fig. 3. Ferrosphere of Jungar fly ash (SEM, back-scattered electron image on polished sections). (a) Plerosphere. (b) A ferrosphere with dendritic magnetite crystallites, the marked area in A is enlarged. (c) A ferrosphere with skeletal magnetite crystallites, the marked area in A is enlarged. (d) Cenosphere. (e) Porous ferrosphere. (f) Dense ferrosphere. (g) Magnetite crystal particles.

This may be attributed to the larger particle size of dry fly ash (average particle size 67–92 μm) in comparison with that of wet fly ash (average particle size 35–66 μm). Finer particles of fly ash have larger surface area, which allows for deposition/absorption of volatile elements on the surface of the fly ash. In addition, some studies (Sakulpitakphon et al., 2000; Mastalerz et al., 2004; Mardon and Hower, 2004) indicated that the content of most semi-volatile elements of fly ash increases with increasing distance from the boiler.

In order to describe the elemental content of combustion residues from the Jungar Power Plant, the following indices were used in this study:

(1) The index f/b = (elemental content in fly ash)/(elemental content in bottom ash) is used to detect elements that are volatilized and subsequently condensed totally or partially in the flue gas (high f/b ratio) or heavy and low-volatile elements that preferably are enriched in bottom ash (Meij, 1994).
(2) The relative enrichment index $Re = ([X]_a/[X]_c) \times (A_{ad}/100)$ was used to describe the enrichment degree of elements in fly and bottom ashes (Meij, 1994). $[X]_a$ is the content of elements in fly or bottom ashes, and $[X]_c$ is the content of corresponding elements in coal. A_{ad} is the ash yield of the coal. Re_f and Re_b were relative enrichment indices for fly and bottom ashes, respectively. $Re = 1$ indicates that element is not enriched or depleted in the fly or bottom ashes, but $Re > 1$ or $Re < 1$ indicates that elements is depleted or enriched, respectively.
(3) The index $Kv = (1 - Re_f \times 0.9/95\% + Re_b \times 0.1) \times 100\%$ was used to describe the volatile proportion of elements during combustion. The production ratio of fly ash to bottom ash in the Jungar Powder Plant was 9:1. The efficiency of the ash collected in the concentration units is 95%.

Table 6 lists all the indices of the wet fly ash in the concentration units.

With the exception of CaO, TiO_2, Fe_2O_3, MgO, Na_2O, Co, Ni, and W, most elements are high in fly ash as compared with the bottom ash ($f/b > 1$). The elements can be classified into three groups in terms of the f/b values, Groups I, II, and III.

Group I. The f/b of Fe_2O_3 and W is <0.8 (0.69 and 0.77, respectively), indicating that Fe_2O_3 and W are much higher in bottom ash than in fly ash. The relative indices Re_f and Re_b of Fe_2O_3 are lowest with exceptions of Hg and Se (0.23 and 0.34, respectively), but those of W are highest (8.87 and 18.1, respectively).

Group II. The oxides and elements with f/b values ranging from 0.8 to 1 include CaO, TiO_2, MgO, Na_2O, MnO, Li, Cr, Co, Ni, and Re. They are evenly distributed in fly and bottom ashes. However, the Re_f and Re_b of TiO_2, MnO, Li, and Cr are less than 1, indicating that both fly and bottom ashes are depleted in these elements. Other elements in this group, with the exception of Re that is a little low in fly ash, show enrichment in fly and bottom ashes.

Group III. The remaining oxides and elements have f/b values higher than 1.2, indicating that all the remaining elements are enriched in fly ash as compared to the bottom ashes. With the exception of Al_2O_3, Ge, Nb, Cs, and Ba whose Re_f is not less than 1, Re_f and Re_b of other elements are <1, indicating that both fly and bottom ashes are depleted in these elements.

In terms of volatile proportion (Kv) of elements during coal combustion, four groups were classified as well, Groups A, B, C, and D.

The oxides and trace elements in Group A have Kv values less than 20%. These oxides and trace elements include Al_2O_3, SiO_2, CaO, MgO, Na_2O, Li, Be, Sc, V, Cr, Co, Ni, Cu, Ga, Ge, Rb, Sr, Y, Zr, Nb, Sn, Sb, Cs, Ba, REEs (rare earth elements), Hf, Ta, W, Re, Th, and U. These elements did not volatilize or a little volatilized during coal combustion.

Much attention should be paid to Ga, because it shows strong enrichment in feed coals and the fly ash and can be potentially extracted and utilized (Dai et al., 2008). In general, it is hard for Ga to volatilize during coal combustion due to its high boiling point. However, the carriers of Ga in the Jungar coals are mainly boehmite and organic matter (Dai et al., 2006a,b, 2008) and part of Ga may form volatile gallium compounds. About 20% of Ga is volatilized after the coal was ashed at 1200 °C (Zhang et al., 2008). The Kv of Ga in this study is 16.5%, similar with the data reported by Zhang et al. (2008).

Some studies indicated that REEs rarely volatilize during coal combustion and are evenly distribute in fly and bottom ashes (Llorens

Table 7
Elements and their enrichment factors in magnetic, MCQ, and glass phases of fly ash from Jungar Power Plant.

Elements	Total fly ash (%)	Magnetic fraction		MCQ fraction		Glass	
		Content (%)	EF	Content (%)	EF	Content (%)	EF
Al_2O_3	49.88	22.95	0.46	66.84	1.34	37.33	0.75
SiO_2	41.74	23.27	0.56	23.61	0.57	56.71	1.36
CaO	2.03	3.54	1.74	0.39	0.19	3.28	1.62
K_2O	0.41	0.16	0.39	0.04	0.1	0.71	1.73
TiO_2	0.67	0.9	1.35	1.23	1.84	0.22	0.33
Fe_2O_3	0.8	39.14	48.93	0.67	0.84	0	0
MgO	0.17	0.36	2.09	0.01	0.06	0.29	1.71
Na_2O	0.12	0.06	0.46	0.01	0.08	0.21	1.74
MnO	0.012	0.108	9.01	0.003	0.21	0.016	1.36
P_2O_5	0.079	0.079	1	0.005	0.06	0.14	1.74
Li	408	31.3	0.08	76.4	0.19	682	1.67
Be	5.23	1.75	0.33	3.36	0.64	6.83	1.31
Sc	17.0	11.1	0.65	10.6	0.63	22.2	1.31
V	68.3	99.2	1.45	69.7	1.02	66.1	0.97
Cr	90.2	80	0.89	36.3	0.4	133	1.48
Co	2.92	13.7	4.69	1.44	0.49	3.74	1.28
Ni	12.3	33.2	2.71	12.8	1.04	11.1	0.91
Cu	37.7	37.3	0.99	19.1	0.51	52.4	1.39
Zn	48.1	94.3	1.96	20.8	0.43	68.1	1.42
Ga	43.4	13.2	0.3	49	1.13	39.9	0.92
As	1.12	11.8	10.5	6.02	5.36	0	0
Se	3.64	0.21	0.06	0.63	0.17	6.13	1.69
Rb	8.49	84.9	10	1.34	0.16	11.6	1.37
Sr	346.5	186	0.54	54.6	0.16	582.9	1.68
Y	34.3	24.7	0.72	29	0.85	38.7	1.13
Zr	713	367	0.52	429	0.6	948	1.33
Nb	81.9	14.5	0.18	44.6	0.54	114	1.39
Mo	3.51	0.75	0.21	3.29	0.94	3.77	1.07
Cd	0.1	0.29	2.94	0.07	0.72	0.11	1.16
In	0.12	0.05	0.44	0.03	0.27	0.2	1.59
Sn	2.7	2.53	0.94	2.35	0.87	2.97	1.1
Sb	0.45	0.97	2.18	0.24	0.53	0.6	1.34
Cs	0.69	5.58	8.15	0.1	0.15	0.99	1.44
Ba	150	428	2.85	140	0.93	149	0.99
La	62.2	43	0.69	70.9	1.14	55.94	0.9
Ce	107	81.8	0.76	91.4	0.85	120.1	1.12
Pr	12.4	9.99	0.81	10.2	0.82	14.2	1.15
Nd	43.4	37.7	0.87	36.9	0.85	48.7	1.12
Sm	8.02	7.4	0.92	7.19	0.9	8.7	1.08
Eu	1.43	1.56	1.09	1.26	0.88	1.56	1.09
Gd	7.2	6.65	0.92	5.82	0.81	8.31	1.15
Tb	1.17	0.87	0.74	0.87	0.75	1.41	1.21
Dy	7.21	5.18	0.72	4.95	0.69	9.06	1.26
Ho	1.44	1.01	0.7	0.92	0.64	1.87	1.3
Er	4.07	2.91	0.72	2.75	0.68	5.14	1.27
Tm	0.6	0.43	0.71	0.43	0.71	0.74	1.24
Yb	4.07	2.85	0.7	2.93	0.72	5.01	1.23
Lu	0.58	0.4	0.68	0.44	0.76	0.7	1.2
Hf	24.4	7.4	0.31	17.9	0.74	30	1.23
Ta	4.23	0.94	0.22	4.14	0.98	4.4	1.04
W	4.82	2.01	0.42	4.19	0.87	5.41	1.12
Re	0.004	0.001	0.25	0.003	0.75	0.005	1.22
Tl	0.47	0.49	1.04	0.04	0.08	0.81	1.72
Pb	55.5	25.8	0.47	22	0.4	82.9	1.5
Bi	0.89	0.51	0.57	1.1	1.24	0.74	0.83
Th	37.0	13.6	0.37	56.9	1.54	21.9	0.59
U	10.6	3.07	0.29	6.37	0.6	14.2	1.34

EF, enrichment factor.

et al., 2001; Wang et al., 2003). Studies by Li et al. (2005) and Sun et al. (2001) showed that bottom ash is more enriched in REEs than fly ash. The Kv value of REEs from Jungar coal combustion residues varies from 10.7 to 16.6%. In addition, they are higher in fly ash than in bottom ash ($f/b = 1.53–1.63$). The differences of the distribution and volatilization of rare earth elements in Jungar combustion residues to previous studies (Llorens et al., 2001; Sun et al., 2001; Wang et al., 2003; Li et al., 2005) were attributed to the occurrence modes of REEs. Generally, REEs in coal are concentrated to clay minerals (Sun et al., 2007; Ren et al., 2006). However, rare earth elements in Jungar coals occurred both in boehmite and in organic matter (Dai et al., 2006a, 2008). The organic-bound REEs could partly volatilize and condense on the fine particles of fly ash, leading to a higher REEs content in fly ash than in bottom ash. Whether REEs are partly volatilized or not may be tested by etching the fly ash to see if they are concentrated on the surface of fly ash. An alternative explanation is that REEs and other elements such as Al in boehmite and organic matter may form extremely small particles when boehmite decomposes and organic matter is burnt out during coal combustion. Such small particles tend to concentrated in finer fractions of fly ash.

Group B includes K_2O, MnO, P_2O_5, Mo, Ag, Cd, In, Au, Tl, Pb, and Bi. These elements have Kv values ranging from 20 to 50%, indicating that

Table 8
Element content and enrichment factors in six size fractions of the fly ash from Jungar Power Plant (The content of oxides and trace elements is in % and μg/g, respectively).

Elements	<120		120–160		160–300		300–360		360–500		>500	
	Content	EF	Content	EF	Content	EF	Content	EF	Content	EF	Content	EF
Al_2O_3	43.84	1	45.98	1	48.33	1	49.12	1	51.93	1	52.22	1
SiO_2	36.91	1.01	39.75	1.03	38.12	0.94	36.22	0.88	33.78	0.78	33.13	0.76
CaO	1.23	0.69	1.68	0.9	2.2	1.12	2.82	1.41	3.71	1.76	3.76	1.77
K_2O	0.25	0.69	0.32	0.85	0.36	0.91	0.38	0.94	0.36	0.84	0.37	0.86
TiO_2	1.12	1.9	1.27	2.06	1.33	2.05	1.57	2.38	1.71	2.45	1.92	2.74
Fe_2O_3	1.52	2.16	1.62	2.2	2.12	2.73	2.48	3.15	2.52	3.03	3.52	4.2
MgO	0	0	0.032	0.2	0.053	0.32	0.043	0.26	0.059	0.33	0.089	0.5
Na_2O	0.18	1.71	0.17	1.54	0.17	1.46	0.18	1.52	0.2	1.6	0.19	1.51
MnO	0.013	1.23	0.013	1.18	0.015	1.29	0.017	1.44	0.019	1.52	0.022	1.75
P_2O_5	0.075	1.08	0.076	1.04	0.082	1.07	0.091	1.17	0.1	1.22	0.12	1.45
Li	280	0.78	348	0.93	326	0.83	327	0.81	344	0.81	335	0.79
Be	4.06	0.88	3.35	0.69	5.2	1.03	5.78	1.12	7.43	1.36	8.19	1.5
F	474	1.04	535	1.12	598	1.19	789	1.55	684	1.27	1162	2.15
Sc	13.6	0.91	15.7	1	19.8	1.21	23.7	1.42	25.4	1.44	29.4	1.66
V	51.3	0.86	57.8	0.92	64.1	0.97	88.3	1.31	103	1.45	109	1.53
Cr	16	0.2	19.9	0.24	22.2	0.25	28.8	0.32	37.3	0.4	42.6	0.45
Co	1.63	0.64	1.74	0.65	2.15	0.76	2.9	1.01	3.61	1.19	4.44	1.45
Ni	6.46	0.6	8.14	0.72	8.87	0.75	9.97	0.83	10.6	0.83	13.1	1.02
Cu	28.3	0.85	46.6	1.34	42.4	1.16	48.9	1.32	59.9	1.53	70.6	1.79
Zn	26.2	0.62	54.4	1.23	40.4	0.87	50.3	1.06	67.5	1.35	86.8	1.73
Ga	25.2	0.66	34.2	0.86	34.4	0.82	41.1	0.96	49.5	1.1	61.2	1.35
As	0.18	0.18	0.33	0.32	0.71	0.66	1.01	0.91	0.42	0.36	1.38	1.17
Se	2.00	0.63	2.29	0.68	2.84	0.81	3.89	1.09	5.33	1.41	5.94	1.56
Rb	5.55	0.74	9.66	1.24	10.9	1.33	10.9	1.3	10.7	1.21	10.8	1.22
Sr	351	1.15	298	0.93	409	1.22	564	1.65	651	1.8	794	2.19
Y	30.7	1.02	34.8	1.1	43.3	1.3	52.8	1.57	57.5	1.61	65.5	1.83
Zr	479	0.76	479	0.73	665	0.96	811	1.16	959	1.29	1077	1.44
Nb	43	0.6	61.5	0.82	70.2	0.89	76.8	0.95	81.6	0.96	83	0.97
Mo	3.59	1.17	3.1	0.96	3.38	1	4.05	1.17	5.18	1.42	5.67	1.55
Cd	0.12	1.38	0.15	1.62	0.16	1.65	0.18	1.93	0.26	2.58	0.34	3.35
In	0.044	0.4	0.075	0.66	0.095	0.79	0.136	1.11	0.183	1.42	0.237	1.83
Sn	3.65	1.54	4.37	1.76	4.05	1.55	5.66	2.13	6.67	2.38	8.62	3.06
Sb	0.35	0.89	0.26	0.63	0.36	0.82	0.52	1.17	0.72	1.55	0.92	1.97
Cs	0.46	0.77	0.82	1.3	0.89	1.33	0.80	1.18	0.71	1	0.74	1.03
Ba	73.9	0.56	166	1.2	150	1.03	141	0.95	150	0.96	150	0.95
La	60.1	1.1	57.9	1.01	77.9	1.29	103	1.68	121	1.87	136	2.09
Ce	104	1.11	104	1.05	112	1.08	167	1.59	206	1.85	246	2.2
Pr	11.7	1.07	11.8	1.03	15.8	1.32	21	1.72	24.6	1.91	28.5	2.2
Nd	39.1	1.03	40.2	1	53.7	1.28	71.4	1.67	84.7	1.87	102	2.24
Sm	7.13	1.01	7.48	1.01	9.88	1.27	12.9	1.63	14.6	1.75	17.2	2.05
Eu	1.25	0.99	1.37	1.04	1.76	1.27	2.31	1.64	2.6	1.75	3.11	2.08
Gd	6.4	1.01	6.85	1.03	9.37	1.34	11.9	1.68	13.3	1.77	15.4	2.04
Tb	1.02	0.99	1.15	1.07	1.48	1.31	1.85	1.61	2.02	1.66	2.33	1.9
Dy	5.99	0.95	6.86	1.03	8.66	1.24	10.4	1.47	11.2	1.49	13	1.72
Ho	1.24	0.98	1.4	1.05	1.76	1.26	2.11	1.49	2.32	1.55	2.66	1.76
Er	3.6	1.01	4.04	1.08	5	1.27	5.9	1.47	6.29	1.49	7.53	1.77
Tm	0.53	0.99	0.59	1.06	0.70	1.21	0.85	1.44	0.90	1.44	1.04	1.65
Yb	3.39	0.95	3.69	0.98	4.74	1.2	5.42	1.35	5.89	1.39	6.88	1.61
Lu	0.50	0.98	0.55	1.02	0.69	1.23	0.79	1.37	0.84	1.39	0.98	1.61
Hf	14.2	0.66	16.4	0.73	20.3	0.86	23	0.96	26.6	1.05	29.5	1.16
Ta	2.85	0.77	4.04	1.04	4.51	1.1	4.91	1.18	5.07	1.15	5.26	1.19
W	3.3	0.78	4.03	0.91	3.9	0.84	3.98	0.84	4.37	0.87	4.55	0.9
Re	0.008	2.3	0.003	0.81	0.003	0.77	0.004	1.02	0.006	1.44	0.005	1.19
Hg	0.120	2.85	0.076	1.72	0.134	2.9	0.100	2.12	0.126	2.53	0.09	1.79
Tl	0.18	0.43	0.31	0.71	0.45	0.99	0.54	1.16	0.65	1.33	0.8	1.63
Pb	16	0.33	30.3	0.59	45.3	0.84	63.5	1.16	81.2	1.41	97.8	1.68
Bi	0.21	0.27	0.40	0.48	0.61	0.71	0.91	1.04	1.27	1.37	1.61	1.73
Th	36.6	1.13	37.2	1.09	44.5	1.24	53.8	1.48	59.4	1.54	66.9	1.73
U	7.89	0.85	9.02	0.92	10.5	1.02	10.9	1.04	12.3	1.11	12.7	1.14

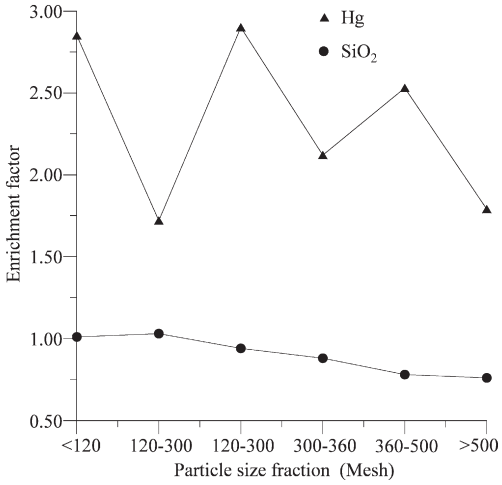


Fig. 4. Variation of enrichment factors of the elements in Type 1.

certain portions of them are volatilized during coal combustion. Although Pb has a high boiling point (1750 °C), it may form volatile lead compounds such as Pb-chlorides with a relatively low boiling point that can volatilize during coal combustion.

Group C includes TiO_2, Fe_2O_3, P_2O_5, F, Zn, and As. They have Kv values ranging from 50 to 80%, indicating that a relatively large proportion of them volatilized during coal combustion. With exceptions of Fe_2O_3 and TiO_2, the Re_f values of the other elements are higher than Re_b values. Elements F and As would condense in the inner part of wet ash concentration unit, leading to much higher Re_f values than Re_b values. Arsenic could easily volatilize and about half of it is emitted into the atmosphere and a small proportion condensed on the fine particles of fly ash. The evaporation of F in this study is different from that reported by Meij and te Winkel (2007), who classified F as an easily-vaporized element in coal from The Netherlands.

Group D includes elements Se and Hg. They have very high Kv values, 83.8 and 86.1%, respectively, indicating that almost all Se and Hg are volatilized during coal combustion. The Re_f and Re_b of Hg and Se in both fly ash and bottom ash were very low (Table 6). A study by Finkelman et al. (1990) indicated that Hg begins to volatilize at a low temperature of 150 °C, and 40–75% Hg is volatilized under 550 °C. Rizeq et al. (1994) found that almost all Hg volatilized under 800 °C. Selenium mainly occurs in galena, clausthalite, and selenio-galena (Dai et al., 2006a). These Se-bearing minerals can be easily decomposed in boiler and then Se is released. Meij and te Winkel (2007) also showed Hg and Se are generally present in ash to a small degree and the largest part is present in the vapor phase.

3.3.2. Elemental composition of different phases of fly ash

As discussed above, we separated the fly ash into three fractions: magnetic, MCQ and glass. The element contents in the magnetic and MCQ fractions were determined by the methods as described in Methods section. The content of elements in the glass may be deduced from the following mass balance equation:

$$C = gC_g + mgC_{mg} + mC_m$$

where C, C_g, C_{mg}, and C_m are the content of element in pristine fly ash, magnetic, glass, and MCQ fractions, respectively. And the g, mg, and m are the proportions of magnetic, glass, and MCQ fractions in the pristine fly ash.

Table 7 presents elemental contents and their enrichment factors of magnetic, MCQ and glass phases of fly ash from the Jungar Power Plant.

The magnetic fly ash contains higher Fe_2O_3 (39.14%) and MnO (0.108%) but lower Al_2O_3 (22.95%) and SiO_2 (23.27%) than MCQ and glass fractions. Oxides CaO, TiO_2, and MgO were high in the magnetic fraction as well, with EF higher than 1.3. With the exception of Al_2O_3 (EF = 1.34) and TiO_2 (EF = 1.84), the major elements in MCQ fraction are low. Glass is enriched in most major elements, except Al_2O_3 and Fe_2O_3.

Most trace elements with EF less than 1 in the magnetic fraction were depleted. The enriched trace elements in magnetic fraction include As, Rb, Cs, Co, Cd, Ba, Ni, Sb, Zn, and V (in order of EF). Most of these elements are lithophiles (Rb, Cs, Ba) or chalcophile (As, Sb, Zn, Cd). It should be noted that As is highly enriched in magnetic fraction with EF up to 10.5. Li et al. (2005) found that As is high in Fe-oxide and Si-Al-Fe-O-rich glass but poor in Si-Al glass. The trace elements with EF < 0.5 in magnetic fly ash include Li, Be, Ga, Se, Nb, Mo, Hf, Ta, W, Pb, Th, and U. Most of these elements are lithophiles.

Trace elements As, Bi and Th are enriched but the content of the remaining elements in mullite are much lower or close to that of the total fly ash (Table 7). Most of the trace elements, with the exception of V, Ni, Ga, As, Bi, and Th, are enriched in the glass.

The REEs are enriched in the glass but depleted in magnetic and MCQ fractions. As compared with light REEs with EF < 1.2, heavy REEs in glass phase are more enriched (EF > 1.2, with an exception of Gd). However, light REEs are more enriched in MCQ and magnetic fractions than heavy REEs.

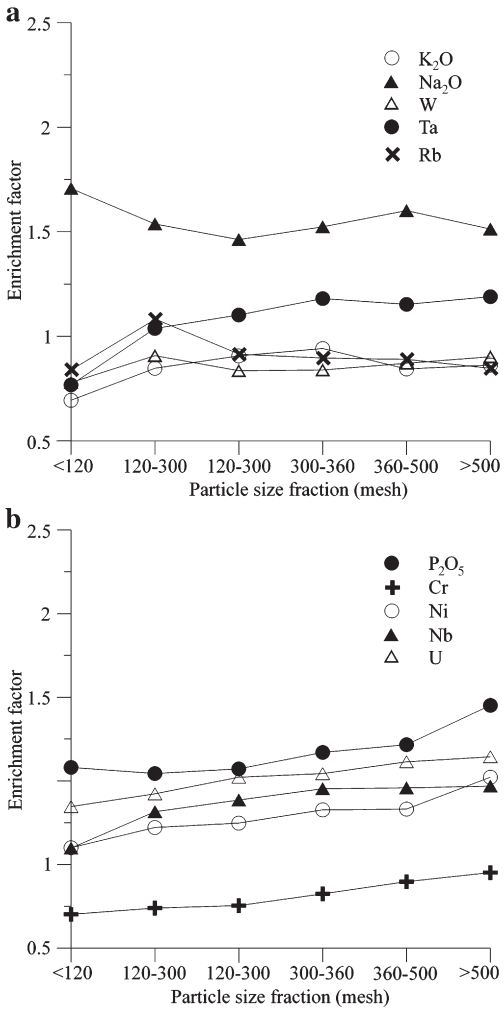


Fig. 5. Variation of enrichment factors of the elements in Type 2. (a) K_2O, Na_2O, W, Ta, and Rb. (b) P_2O_5, Cr, Ni, Nb, and U.

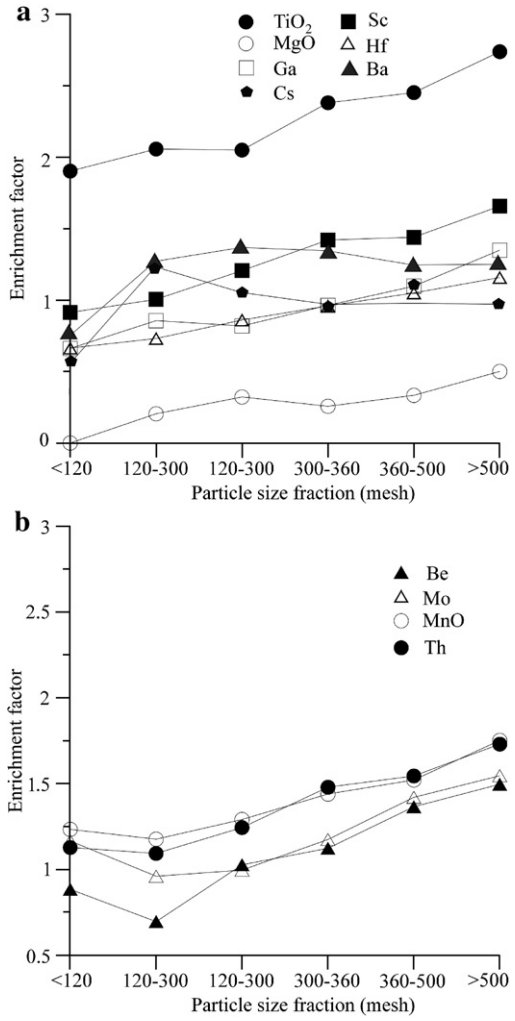


Fig. 6. Variation of enrichment factors of the elements in Type 3. (a) TiO_2, MgO, Ga, Cs, Sc, Hf, and Ba. (b) Elements Be, Mo, MnO, and Th.

3.3.3. Elemental distribution in different particle size fractions of fly ash

Table 8 listed the elemental contents and enrichment factors of six size fractions of dry fly ash. It can be seen that the contents of most elements increase with the decrease of the particle size. This indicates that most elements could more or less volatilize during coal combustion. The volatilized elements condensed on the surface of the fine particles which have a larger surface area than coarse particles. In addition, the low-content accessory minerals in fly ash generally have a size of 0.1–1 μm and were also apt to being enriched on the surface of fine particles (Vassilev and Vassileva, 1996). The accessory minerals are the important carriers of some elements in the fly ash (Vassilev and Vassileva, 1996).

Four types of the elements in fly ash can be separated based on the content variations in different size fractions.

Type 1 includes SiO_2 and Hg. SiO_2 and Hg have a decreasing trend with the decrease of the particle size (Fig. 4). As discussed above, the content of mullite increases but glass which is richer in SiO_2 than mullite, decreases with the decrease of the particle size.

The decreasing of Hg content with decreasing particle size is probably attributed to the decreasing char content with the decrease of particle size. Some studies (Hower et al., 2000; Sakulpitakphon et al., 2000, 2004; Mardon and Hower, 2004) indicated that mercury capture shows a positive correlation with char content of fly ash and a negative correlation with the flue gas temperature. The mercury content is weakly positively correlated to the char content (the correlation coefficient is 0.35, based on six samples) in the Jungar fly ash.

Type 2 includes K_2O, Na_2O, P_2O_5, Nb, Cr, Ta, U, W, Rb, and Ni. The content of these elements did not show a clear trend with the change of particle size (Fig. 5). Most of these elements are enriched either in mullite or in aluminosilicate glass and relatively hard to volatilize during coal combustion. With exceptions of K_2O and P_2O_5, the volatile proportions of elements in Type 2 are less than 20%. The uniform distribution of these elements in different size fractions indicates that there is a weak relationship between the enrichment and surface area. They are probably evenly distributed in the inner part and surface of the fly ash particles.

Type 3 includes elements Ti, Mn, Be, Ga, Sc, Hf, and Th that slightly increase with the decrease of particle size (Fig. 6). With exceptions of Be and Sc with Kv values of 5.9 and 7.4, respectively, those elements have 20–80% Kv values and were included in the Group B. These elements are lithophiles.

Type 4 includes CaO, Fe_2O_3, Se, F, Co, V, Cu, Zn, As, Sr, Zr, Cd, Sn, Sb, Tl, Pb, Bi, and REEs that remarkably increase with the decrease of particle size (Fig. 7). The content of Fe_2O_3 significantly increases with the decrease of particle size. Selenium has a very high Kv value (83.8%), but some elements in Type 4 have very low Kv values (for example, CaO, Co, V, Cu, Sr, and Zr) and some have medium Kv values (For example, Fe_2O_3, F, Zn, As, Cd, Sn, Sb, Tl, Pb, and Bi). The enrichment factors of light REEs become higher than those of heavy ones with decreasing particle size (Fig. 7g).

4. Conclusions

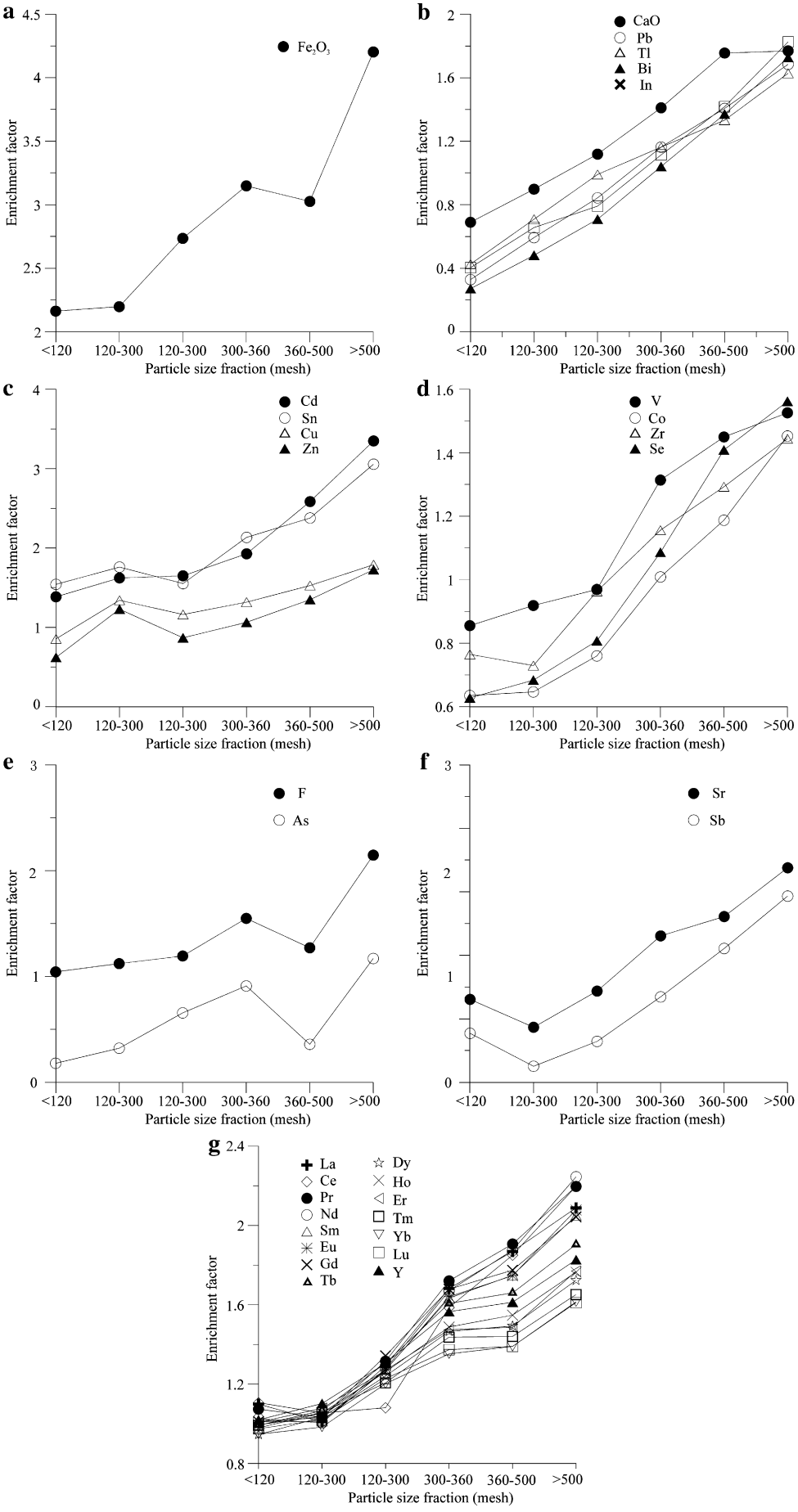
The mineralogical composition of fly ash from the Jungar Power Plant is controlled largely by mineral matter in feed coal. The bottom and fly ashes from the economizer and wet concentration units mainly consist of amorphous glass, mullite, corundum, quartz, char, and minor amounts of calcite and K-feldspar.

Glass is the predominant component in fly ash (52.6%–54.8%) and bottom ash (54.4%). Mullite is the most abundant mineral found in high-alumina fly ash (34.9%–37.4%) and bottom ash (27.2%). It crystallizes from aluminosilicate melt that is derived from kaolinite in feed coal. Corundum, the characteristic mineral in high-alumina fly ash from Jungar Power Plant, is the second most abundant mineral in the alumina-rich fly ash (3.9%–4.0%) and bottom ash (3.2%). It crystallizes from alumina melt derived from boehmite.

The glass abundance in fly ash decreases with decreasing particle size. To the contrary, corundum increases with decreasing particle size, and it reaches 10.9% in the <500 mesh size fraction. The 300–360-mesh fraction is slightly higher in mullite (37.5%) than other size fractions (32.3%–35.9%).

The fly ash from the Jungar Power Plant is characterized by the high Al_2O_3 content, with an average of 51.9%. However, the fly ash is poor in SiO_2, Fe_2O_3, CaO, MgO, Na_2O, P_2O_5, and As. With the exception of CaO, TiO_2, Fe_2O_3, MgO, Na_2O, Co, Ni and W, most elements are enriched in fly ash as compared with the bottom ash. The elements can be classified into three groups in terms of the f/b ratio. Groups I include Fe_2O_3 and W with $f/b < 0.8$. The oxides and elements in Group II (CaO, TiO_2, MgO, Na_2O, MnO, Li, Cr, Co, Ni, and Re) have f/b values ranging from 0.8 to 1. The remaining oxides and elements in Group III have f/b values higher than 1.2, indicating that all the remaining elements are enriched in fly ash as compared to the bottom ashes.

In terms of volatile proportion (Kv) of the elements during coal combustion, four groups (A, B, C, and D) were classified. The oxides or elements in Group A (Al_2O_3, SiO_2, CaO, MgO, Na_2O, Li, Be, Sc, V, Cr, Co, Ni, Cu, Ga, Ge, Rb, Sr, Y, Zr, Nb, Sn, Sb, Cs, Ba, REEs, Hf, Ta, W, Re, Th, and U) have Kv values less than 20%. Group B (including K_2O, MnO, P_2O_5, Mo, Ag, Cd, In, Au, Tl, Pb, and Bi) has Kv values ranging from 20 to 50%. The oxides or elements including TiO_2, Fe_2O_3, P_2O_5, F, Zn, and As in


a
Fe2O3
Enrichment factor
Particle size fraction (mesh)
<120
120-300
120-300
300-360
360-500
>500
b
CaO
Pb
Tl
Bi
In
c
Cd
Sn
Cu
Zn
d
V
Co
Zr
Se
e
F
As
f
Sr
Sb
g
La
Ce
Pr
Nd
Sm
Eu
Gd
Tb
Dy
Ho
Er
Tm
Yb
Lu
Y

Group C have Kv values ranging from 50 to 80%, indicating that a relatively large proportion of these elements are volatized during coal combustion. Group D includes elements Se and Hg, having very high Kv values (>80%) indicating that almost all Se and Hg are volatilized during coal combustion.

Arsenic, TiO_2, Th, Al_2O_3, Bi, La, Ga, Ni, and V are high in mullite, and the magnetic matter is enriched in Fe_2O_3, CaO, MnO, TiO_2, Cs, Co, As, Cd, Ba, Ni, Sb, MgO, Zn, and V. The remaining elements are high in the glass fraction. The abundances of K_2O, Na_2O, P_2O_5, Nb, Cr, Ta, U, W, Rb, and Ni do not have a clear trend with particle size changing, while SiO_2 and Hg decrease and the remaining elements clearly increase with decreasing particle size.

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Fig. 7. Variation of enrichment factors of the elements in Type 3. (a) Fe_2O_3. (b) CaO, Pb, Tl, Bi, and In. (c) Elements Cd, Sn, Cu, and Zn. (d) Elements V, Co, Zr, and Se. (e) Elements F and As. (f) Elements Sr and Sb. (g) Rare earth elements.

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